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The smart future for sustainable development: Artificial intelligence solutions for sustainable urbanization

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Abstract

Future tools for supporting collaborations between technology and sustainable development include artificial intelligence (AI) applications in sustainable Urbanization roles. This article highlights the various applications of AI in advancing sustainable urbanization. From urban planning to disaster management, AI technology is revolutionizing the way cities are designed and managed. By leveraging data analytics, machine learning, and predictive modeling, AI is helping city officials make informed decisions, optimize resource usage, and improve quality of life for urban residents. Despite the immense potential of AI in sustainable urban development, there are still challenges and limitations to overcome. We show some of the most significant problems related to these issues. These include issues related to data privacy, algorithm bias, and ethical considerations. Continued research and innovation are needed to address these challenges and ensure that AI technology is used responsibly and effectively in shaping sustainable cities. As a result, AI has the power to transform urban environments and create more sustainable, resilient communities. By harnessing the capabilities of AI, cities can become more efficient, environmentally-friendly, and prepared for the challenges of the future. It is essential for policymakers, urban planners, and technology developers to work together to harness the full potential of AI in sustainable urbanization and create a better future for all. Proactively addressing these challenges can unlock the full potential of AI in combating sustainable cities and building a sustainable future for all.

KEYWORDS
artificial intelligence, collaboration, energy efficiency, predictive modeling, sustainable cities, sustainable development goal 11

1 | INTRODUCTION

Urbanization is a prevalent trend in modern society, with more people moving to urban areas seeking better opportunities and quality of life. This migration has brought challenges such as pollution, traffic, and infrastructure issues (Adebayo & Ullah, 2024; Bello et al., 2024; Yeyouomo & Asongu, 2024). To ensure the sustainability of cities, innovative solutions are necessary. One new idea that is becoming more popular in the field of sustainable urban development is the use of artificial intelligence (AI) (d'Orville, 2020; Leal Filho et al., 2022; Pee & Pan, 2022). AI, which involves machines being programmed to think and learn like humans, has the potential to greatly change urban planning and development. By using AI technology, city planners and policymakers can access valuable information and data-driven analyses to make informed decisions that support efficient, sustainable, and livable cities (Al-Raeei, 2023a; Singh et al., 2024; Truby, 2020; Zhao, Song, et al., 2023). The integration of AI in urban planning offers many benefits, from improved resource distribution to better environmental protection. One of the main advantages of AI is its ability to process large amounts of data quickly, allowing city officials to identify trends and patterns that may not have been noticed otherwise. This data-driven approach can help with decisions related to infrastructure, transportation, and waste management, leading to more effective and sustainable urban solutions. AI technology can support the implementation of smart city initiatives by utilizing interconnected devices and sensors to enhance city functions and elevate quality of life. Through the analysis of real-time data from these devices, AI systems can pinpoint inefficiencies, project future trends, and streamline processes related to energy, transportation, and safety, ultimately resulting in cost savings, reduced environmental impact, and enhanced living conditions in urban areas.

Moreover, AI can aid in tackling environmental issues in cities, such as pollution. By employing AI-driven monitoring systems, city officials can monitor pollution levels, pinpoint sources of contamination, and take targeted actions to safeguard the environment. AI technology can also assist in crafting models to prepare for climate change and natural disasters, enabling cities to brace themselves for extreme weather events. In addition to these practical applications, AI has the potential to boost citizen engagement in urban planning. By utilizing AI platforms and tools, city governments can collect feedback, inspire innovation, and engage residents in decision-making processes regarding urban development projects, fostering transparency, accountability, and cooperation among members of the community. However, the use of AI in urban planning poses challenges and ethical considerations that must be taken into account. AI systems can exhibit biases and discriminations, particularly in areas such as housing and transportation. It is crucial for city officials to prioritize equity and inclusivity when implementing AI technologies to ensure that all community members reap the benefits of advancements in urban planning. Furthermore, concerns surrounding data privacy, cybersecurity, and potential job displacement must be addressed when incorporating AI into urbanization efforts. City governments must safeguard personal data, establish robust cybersecurity protocols, and provide training programs to help individuals adapt to an evolving automated landscape. Policymakers should collaborate with experts and organizations to devise guidelines and regulations for the responsible application of AI in urban planning. In summary, integrating AI into sustainable urban development holds the promise of revolutionizing cities for the better. By harnessing AI technology, city planners and policymakers can make informed decisions that maximize resources, enhance efficiency, and elevate quality of life for urban dwellers. Nonetheless, it is imperative to confront ethical concerns, prioritize inclusivity, and foster collaboration among all community members to fully realize the benefits of AI in urban planning. Through careful planning and effective execution, AI has the potential to propel cities towards a more sustainable and prosperous future. Additionally, AI has the capacity to enhance waste management practices by optimizing collection routes and identifying opportunities for recycling and waste reduction. In the agricultural sector, AI tools can play a critical role in refining crop management practices and mitigating the environmental impact of farming activities. By scrutinizing data on soil quality, weather patterns, and crop health, AI algorithms can guide farmers in making better decisions regarding planting, irrigating, and harvesting crops, resulting in amplified crop yields and decreased resource consumption. Moreover, AI can support conservation efforts and wildlife preservation by tracking endangered species, monitoring their populations, and detecting illicit activities like poaching and deforestation. Through the utilization of AI-powered drones, cameras, and sensors, conservationists can access real-time data on wildlife habitats and enact proactive measures to safeguard vulnerable species and ecosystems. In the fight against climate change, AI has the potential to revolutionize our approach to developing climate models, predicting extreme weather events, and optimizing renewable energy production (Al-Raeei, 2023b; Cotton & Winter, 2022; Kahupi et al., 2023; Pandey et al., 2021; Rweyendela, 2022; Shenkoya & Kim, 2023; Stevenson et al., 2021; Thiel, 2019; Yoo & Jeon, 2022; Zhang et al., 2023). AI algorithms can play a key role in advancing the understanding of the complex connections between climate factors and predicting the impacts of climate change more accurately. Additionally, AI can help optimize the production and distribution of sustainable energy sources such as solar and wind power, leading to the creation of a more environmentally friendly and resilient energy infrastructure. The effective management of supply chains is another area where AI can make a significant impact. By utilizing AI technologies, businesses can streamline their supply chains, reduce waste, and minimize their environmental footprint. AI allows companies to track product movements, identify opportunities for resource efficiency, and decrease emissions from transportation and manufacturing activities. In addition to its environmental benefits, AI also contributes to the achievement of the Sustainable Development Goals by promoting sustainable economic growth, creating new job opportunities, and enhancing decision-making processes (Abad-Segura et al., 2020; Al-Raeei, 2023c; Al-Raeei, 2024; Awajan, 2022; Gawel et al., 2022; Ruyffelaert, 2022; Yoo & Jeon, 2022). Utilizing AI technologies like machine learning and predictive analytics, cities can maximize their resources, enhance public services, and elevate residents' quality of life. AI enables access to education, healthcare, renewable energy, and disaster relief, fostering a more sustainable and inclusive global community. As such, AI plays a crucial role in advancing sustainable urban development (Al-Raeei et al., 2023; Al-Raeei et al., 2024; Bustamante et al., 2022; Ensslin et al., 2022; Hossain et al., 2022; Muro et al., 2020; Sandanayake et al., 2022; UNICEF MENA, 2015; Zhao, Zhang, et al., 2023). AI can also play very significant role in all of sustainable development goals (Figure 1) supported by the United Nations.

The use of AI technology holds promise in transforming how cities tackle environmental issues, providing avenues to enhance awareness of climate change, create predictive models, track environmental information, enhance energy conservation, optimize city planning, and bolster strategies for adapting to and building resilience against climate challenges which supports the sustainable urbanization by AI.

In this article, we discuss the key applications of AI techniques in supporting sustainable urbanization (see Figure 2). The second section focuses on the application of AI in urban planning, highlighting its role in shaping cities for the future. Moving on to the third section, we delve into how AI can assist in resource and traffic management to create more sustainable urban environments. The fourth

FIGURE 1 The SDGs.

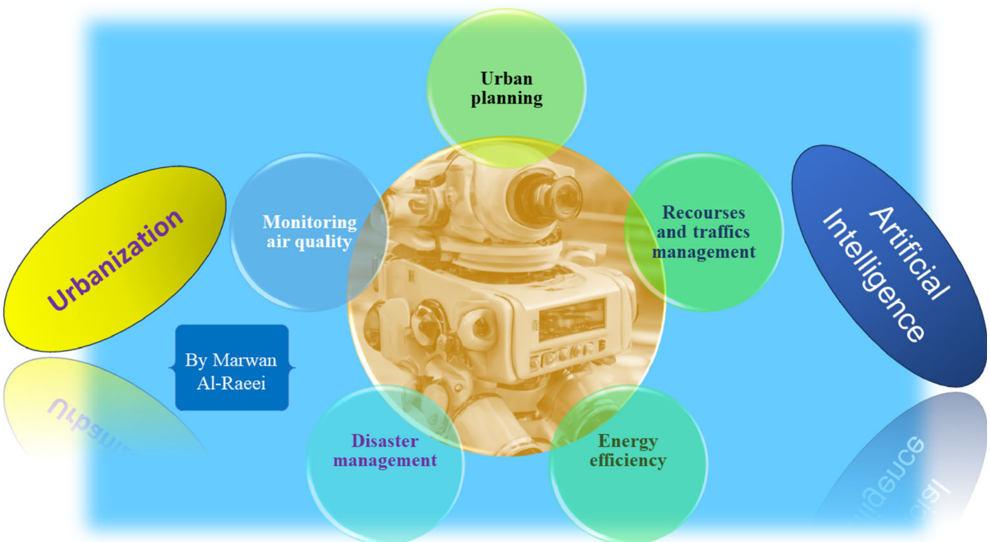

FIGURE 2 The applications of artificial intelligence for sustainable urbanization.

section explores the monitoring of air quality in sustainable urbanization through the use of AI, showcasing the potential for improving public health and well-being. Energy efficiency in urban development is then tackled in the fifth section, emphasizing the importance of AI in achieving environmentally-friendly cities. Disaster management is the focus of the sixth section, underscoring the critical role AI plays in mitigating the impact of natural disasters and ensuring resilient urban communities. Despite the significant benefits of incorporating AI in sustainable urbanization, there are limitations to consider, which are discussed in the seventh section of the article. In last section of the article which is the conclusion, the article wraps up by summarizing the key points discussed and emphasizing the importance of continued research and innovation in leveraging AI for sustainable urbanization.

2 | URBAN PLANNING

Urban planning plays a crucial role in shaping the development of cities to be sustainable, environmentally-friendly, and efficient for the community. With the challenges posed by rapid urbanization and climate change, urban planners must adapt and utilize innovative tools to design cities that can withstand these pressures and support the needs of their residents. In recent years, the integration of AI tools in urban planning has shown promise in aiding planners in making informed and data-driven decisions to create more resilient and sustainable cities (Jha et al., 2021).

The use of AI in urban planning offers a plethora of benefits to urban planners by streamlining the process of analyzing vast amounts of data on various aspects including population density, land use, transportation patterns, energy consumption, and more. Through machine learning algorithms, AI can recognize trends and patterns within the data, enabling planners to have a deeper understanding of the dynamics of urban development and helping them make decisions based on evidence (Sanchez et al., 2023). For instance, AI can predict future population growth in a city and suggest suitable locations for new housing developments to meet the growing demands, thereby preventing urban sprawl and ensuring new developments are situated in areas well-connected to public transportation and amenities. An essential advantage of utilizing AI in urban planning is its ability to optimize land use and enhance efficiency (Yigitcanlar et al., 2020). By analyzing data on land availability, zoning regulations, and environmental restrictions, AI tools can assist planners in identifying underutilized areas that can be developed for housing, commercial or green spaces. This approach can lead to reduced urban congestion and minimize the environmental impact of development by encouraging the creation of denser, mixed-use neighborhoods that are pedestrian-friendly and well-served by public transit. AI also presents opportunities for urban planners to design more sustainable transportation systems that aim to reduce greenhouse gas emissions and improve air quality. By analyzing data on traffic patterns, public transit utilization, and biking infrastructure, AI tools can propose improvements to existing infrastructure such as the implementation of new bus routes or bike lanes that can alleviate congestion and promote alternative modes of transportation. This approach can help cities meet their climate objectives and foster healthier and more liveable environments for urban residents.

Moreover, AI can play a vital role in enhancing disaster resilience and emergency response planning. By examining data on natural hazards such as floods or earthquakes, AI tools can help identify high-risk areas and recommend mitigation strategies to minimize the impact of disasters. This proactive approach enables cities to better prepare for emergencies and safeguard vulnerable populations from harm. In general, the integration of AI in urban planning has the potential to transform the way cities are conceptualized and managed. By harnessing the capabilities of data analytics and machine learning algorithms, urban planners can make informed decisions that prioritize sustainability, resilience, and equity in urban development. As AI technology continues to evolve, it is imperative for urban planners to adapt and incorporate these tools in their planning processes to create cities that are not only environmentally-friendly and efficient but also inclusive and liveable for all residents.

3 | RECOURSES AND TRAFFICS MANAGEMENT

The managements of the resources and traffics using AI play key roles in applying of AI for sustainable Urbanization. Resource management is a critical component of sustainable urban development, particularly in light of the challenges posed by a growing population and increasing urbanization. Cities around the world are grappling with the task of effectively managing resources such as electricity, water, and waste to ensure efficient city operations and promote sustainability. In this regard, AI technology has emerged as a valuable tool for optimizing resource allocation and consumption in urban areas. One of the primary advantages of utilizing AI in resource management is its capacity to analyze vast amounts of data in real-time. By leveraging AI algorithms, city officials can predict patterns of resource consumption based on historical data, weather conditions, and other relevant factors. This enables decision-makers to make informed choices regarding resource allocation and distribution, leading to more efficient resource management practices.

Electricity management stands out as an area where AI technologies can have a profound impact on optimizing resource consumption. Smart grid systems powered by AI have the capability to monitor energy usage, identify anomalies, and adjust supply and demand in real-time. By doing so, these systems can help to minimize wastage, reduce costs, and enhance the overall efficiency of the electricity grid, ultimately leading to a more sustainable energy infrastructure. Water management is another critical aspect of urban resource management where AI can play a significant role. AI technologies can be employed to monitor water quality, detect leaks in the supply network, and optimize water distribution systems. Through the analysis of data collected from sensors and meters, AI algorithms can pinpoint areas of high water consumption and propose conservation strategies, thereby promoting more efficient and sustainable water management practices in urban areas. Waste management also stands to benefit from the integration of AI technologies. Waste management is an essential aspect of modern-day society, as the amount of waste generated continues to increase at an alarming rate. With the integration of AI technologies, waste management processes can be significantly improved to benefit both the environment and the communities in which we live. AI-powered systems have the capability to optimize waste collection routes, identify recycling opportunities, and track waste disposal trends. Through the analysis of data collected from waste bins and sensors, AI algorithms can provide invaluable insights to assist cities in reducing landfill waste, increasing recycling rates, and promoting sustainable waste management practices. This not only helps in minimizing the impact on the environment but also contributes to the overall well-being of the community.

One of the key benefits of using AI technology in waste management is its ability to gather and analyze data on waste generation patterns. By understanding when and where waste is being generated, municipalities can optimize their collection routes for garbage trucks, reducing unnecessary trips and lowering fuel consumption. Additionally, AI can help predict maintenance needs for recycling facilities, ensuring that they remain operational and efficient. Furthermore, AI can play a crucial role in improving recycling processes by sorting, separating, and processing materials more effectively. Traditional recycling facilities often struggle with contamination issues, as different types of materials are mixed together, making it difficult to recover and reuse them. AI technology can help overcome this challenge by accurately sorting materials based on their composition, increasing the amount of materials that can be recycled and decreasing the amount that ends up in landfills. The potential impact of AI on waste management practices is immense. By making these processes more efficient, cost-effective, and environmentally sustainable, cities can drastically reduce their carbon footprint, conserve valuable resources, and create a cleaner, healthier environment for all residents. This transformation is not only essential for meeting sustainability goals and regulatory requirements but also for ensuring a high quality of life for current and future generations. In addition to the environmental benefits, the integration of AI in waste management can also lead to economic advantages for cities and municipalities. By optimizing waste collection routes and recycling processes, operational costs can be reduced, saving taxpayer dollars and improving overall budget planning. Furthermore, the increased efficiency and effectiveness of waste management practices can attract businesses and investors that prioritize sustainability, leading to economic growth and job creation within the community. One of the ways in which AI technology can drive economic growth in the waste management sector is by enabling the development of innovative solutions and services. Startups and companies specializing in AI-driven waste management technologies can create new job opportunities and drive investment in research and development. By fostering a culture of innovation and entrepreneurship, cities can position themselves as leaders in sustainable waste management practices and attract talent and investment from around the world.

Moreover, AI can help cities and municipalities better understand the behavior and preferences of their residents when it comes to waste management. By analyzing trends and patterns in waste generation and disposal, local authorities can tailor their waste management strategies to meet the specific needs of the community. This not only enhances the overall effectiveness of waste management initiatives but also fosters a sense of civic pride and responsibility among residents. Another important aspect of AI in waste management is its potential to improve public health and safety. Proper waste management is essential for preventing the spread of diseases, pests, and contaminants that can pose serious health risks to the community. By using AI to optimize waste collection, recycling, and disposal processes, cities can reduce the likelihood of public health issues arising from improper waste management practices. This in turn leads to a cleaner, safer environment for all residents to live and work in. As a result, the integration of AI technologies in waste management has the potential to revolutionize the industry and create a more sustainable future for our planet. By leveraging the power of data analytics, machine learning, and automation, cities can make informed decisions that lead to more efficient, cost-effective, and environmentally friendly waste management practices. The benefits of using AI in waste management are far-reaching, impacting not only the environment but also the economy, public health, and overall quality of life for residents. As we continue to face pressing challenges related to waste generation and disposal, AI stands as a powerful tool to help us address these issues and build a brighter, more sustainable future for all. In general, AI technology presents cities with a powerful tool for enhancing resource allocation and consumption practices. By harnessing the capabilities of AI, urban areas can strive to minimize waste, conserve valuable resources, and champion sustainable practices that not only benefit the environment but also contribute to the well-being of residents and the broader community.

On the other hand, traffic congestion is a pervasive issue in urban areas, resulting in extended travel times, heightened pollution levels, and increased fuel consumption (Alkinani et al., 2022; Mandal et al., 2020; Sadek et al., 1998; Shaygan et al., 2022; Tang et al., 2022). To address these challenges, cities are increasingly turning to AI-powered systems for traffic management. AI technology can effectively optimize traffic flow, alleviate congestion, and enhance the overall efficiency of the transportation network, ultimately contributing to a more sustainable urban environment. One of the key advantages of utilizing AI in traffic management lies in its ability to analyze real-time traffic data sourced from various channels. AI algorithms can process information from traffic cameras, sensors, and GPS devices to anticipate traffic patterns, pinpoint congestion hotspots, and recommend optimal routing strategies. This real-time analysis enables cities to promptly respond to fluctuating traffic conditions and adjust traffic flow accordingly to alleviate congestion and enhance the overall efficiency of the transportation system. AI-powered systems can also enhance traffic signal timing to reduce wait times at intersections and optimize traffic flow. By scrutinizing traffic data and dynamically adjusting signal timings, cities can curtail congestion, lower emissions, and bolster the effectiveness of the transportation infrastructure, leading to a more sustainable and streamlined urban mobility system. Moreover, the integration of AI technologies with other smart city initiatives can further amplify the benefits of traffic management. For instance, AI-powered systems can communicate with autonomous vehicles to coordinate traffic flow and prevent collisions, thereby enhancing road safety and efficiency. By leveraging AI technology, cities can establish a cohesive and efficient transportation network that prioritizes sustainability, safety, and convenience for residents and visitors alike. In summary, AI technology offers urban areas a potent solution for optimizing traffic management, reducing congestion, and promoting sustainable urban mobility. Through the implementation of AI-powered systems, cities can enhance the efficiency of their transportation networks, mitigate emissions, and cultivate a more sustainable urban landscape that caters to the needs of its inhabitants and fosters a high quality of life.

4 | MONITORING AIR QUALITY

With the growing concern over air pollution and its detrimental effects on public health, the need for effective air quality monitoring in urban areas has never been more pressing. As cities expand and industrial activities increase, the levels of pollutants in the air are on the rise, posing serious risks to the health of residents. The development of innovative solutions, such as AI-powered sensors, offers a promising way to tackle this urgent issue and improve air quality in urban regions (Neo et al., 2023). Air pollution is a complex problem that requires a multifaceted approach to address. Traditional air quality monitoring methods, such as stationary monitoring stations, are limited in their ability to provide timely and accurate data on pollution levels in different parts of a city (Montalvo et al., 2022). These methods often rely on manual operation and periodic sampling, which can result in gaps in data and hinder the ability to effectively track pollution sources and trends.

In contrast, AI-powered sensors offer a more dynamic and efficient way to monitor air quality in real-time. These sensors use AI algorithms to analyze data on pollution levels and trends, enabling city officials to quickly identify sources of pollution and take targeted actions to alleviate the problem. By continuously monitoring air quality at various locations within a city, AI-powered sensors provide a comprehensive and up-to-date picture of the pollution levels, allowing for better decision-making and intervention strategies (William et al., 2023). One of the key advantages of AI-powered sensors is their ability to integrate data from multiple sources, such as weather and traffic information, to analyze the factors influencing air pollution. By examining the relationship between pollution levels and external factors like temperature, humidity, and wind speed, these sensors can help city officials better understand the dynamics of air quality and develop more effective strategies to reduce pollution levels. Furthermore, AI-powered sensors can be strategically deployed in areas with high pollution levels or vulnerable populations, allowing for targeted interventions to improve air quality and protect public health. By combining data from these sensors with information on population density, health indicators, and socioeconomic factors, city officials can develop customized strategies to address the specific needs of different communities and reduce the impact of air pollution on residents. The use of AI-powered sensors in air quality monitoring represents a significant step forward in the fight against air pollution in urban areas. By harnessing the power of AI and real-time data analysis, these sensors have the potential to revolutionize the way cities approach air quality management and mitigate the health risks associated with poor air quality. As AI technology continues to advance and become more accessible, the integration of AI-powered sensors into air quality monitoring systems is likely to become more widespread. By leveraging the capabilities of AI, cities can not only improve the accuracy and efficiency of air quality monitoring but also enhance their ability to protect public health and create a more sustainable urban environment for future generations. In general, air quality monitoring is a critical component of environmental management in urban areas, and the use of AI-powered sensors offers a promising solution to improve the effectiveness of these efforts. By leveraging the capabilities of AI and real-time data analysis, cities can better track pollution levels, identify sources of contamination, and develop targeted interventions to improve air quality and safeguard public health. As cities continue to grow and face increasing challenges related to air pollution, AI-powered sensors will play an essential role in ensuring the health and well-being of urban residents in the years to come.

5 | ENERGY EFFICIENCY

The field of AI for energy efficiency is rapidly progressing, encompassing the use of AI technology to optimize energy consumption and enhance efficiency in various applications within urban areas, such as buildings, lighting systems, and infrastructure. By employing AI algorithms and machine learning techniques, energy management systems can effectively analyze and interpret data from sensors, meters, and monitoring devices to detect energy usage patterns, trends, and anomalies in real-time. This proactive approach enables organizations to make informed, data-driven decisions to minimize waste and enhance overall efficiency in sustainable urbanization efforts. Multiple recent studies have focused on the use of AI in the light of energy. For example, a study examines the impact of AI application on enterprise environmental performance, specifically focusing on microenterprises. The research provides evidence showing the effects of AI on environmental sustainability within small businesses (Shang et al., 2024). Another research explores the role of AI in renewable energy, considering both the benefits and disadvantages associated with its implementation. The study delves into the implications of AI for the renewable energy sector, highlighting its potential impacts on energy economics (Qin et al., 2024). Also, a study investigates the effects of AI on energy transition, with a focus on the moderating role of the Paris Agreement. The research aims to understand how AI influences the shift towards cleaner energy sources and sustainable practices, particularly in relation to global climate goals (Chishti et al., 2024). A research provides new evidence on the relationship between AI and new energy vehicles, examining whether it is a win-win situation for both. The study explores the potential synergies between AI technology and the development of new energy vehicles, aiming to shed light on their mutual benefits (Gu et al., 2024). Additional study delves into how AI can promote renewable energy development, particularly through the facilitation of climate finance. The research examines the role of AI in advancing renewable energy projects and attracting financial support, emphasizing the importance of AI in addressing climate change challenges (Zhao et al., 2024). The benefits of Energy Efficiency AI are extensive and impactful. Real-time insights into energy usage patterns allow for immediate adjustments to optimize consumption, such as automatically regulating lighting levels and heating settings based on factors like occupancy and weather conditions. This automation not only reduces energy waste but also lowers operational costs and enhances user experience in urban environments. Within building management, Energy Efficiency AI presents significant opportunities for facility managers and building owners to enhance energy efficiency and cost savings. By analyzing data on energy consumption, occupancy patterns, and building performance, AI algorithms can pinpoint areas for energy savings and recommend strategies to implement them. This forward-thinking approach not only reduces energy waste but also bolsters the sustainability of buildings, contributing to a greener future for urban areas. Furthermore, Energy Efficiency AI can be incorporated into smart grid systems and energy management platforms to optimize energy distribution and bolster the reliability of the power grid. Through AI technology, utilities and grid operators can improve load forecasting, demand response, and energy storage management, leading to a more efficient and sustainable energy infrastructure. This optimization benefits utility companies by cutting costs and enhancing operational efficiency, while also ensuring consumers have access to dependable and affordable energy services.

By harnessing the potential of AI, Energy Efficiency AI offers promising prospects for organizations to reach their energy efficiency and sustainability objectives. From reducing carbon emissions to lowering energy costs, AI technology provides a holistic approach to enhancing energy management practices and advancing towards a more sustainable future. As organizations prioritize sustainability and environmental responsibility, Energy Efficiency AI will play a pivotal role in spurring innovation and progress within the energy sector. In summary, Energy Efficiency AI is a transformative tool with the power to revolutionize how we manage and consume energy in urban settings. By leveraging AI capabilities, organizations can redefine their energy management strategies, minimize waste, and boost overall efficiency. As we strive towards a sustainable future, the integration of Energy Efficiency AI will be vital in achieving goals of cutting carbon emissions, reducing energy expenses, and forging a greener world for generations to come.

6 | DISASTER MANAGEMENT

Disaster management encompasses a wide range of activities and processes that are aimed at reducing the impact of natural disasters on communities and ensuring a timely and effective response when disasters occur. It involves a complex interplay of planning, coordination, communication, and resource management to protect lives, property, and the environment. One of the key challenges in disaster management is the unpredictability of natural disasters. While we can use historical data and scientific knowledge to understand the probability and potential impacts of certain types of disasters, there is always an element of uncertainty involved. This is where AI technology can be particularly valuable.

AI algorithms have the ability to analyze vast amounts of data from various sources, such as weather patterns, geological data, demographic information, and social media feeds, to identify patterns and relationships that can help predict the likelihood and severity of future disasters. By using predictive modeling techniques, AI can generate forecasts and alert authorities to potential risks, enabling them to take proactive measures to mitigate the impact of disasters. For example, AI-powered systems can analyze the movement of hurricanes and predict their trajectory, intensity, and potential impact on coastal communities. This information can be used to issue timely warnings, evacuate at-risk populations, and mobilize resources such as emergency shelters, medical supplies, and search and rescue teams. Similarly, AI can help predict the likelihood of earthquakes and identify vulnerable areas where building codes may need to be reinforced to minimize damage. In addition to predictive modeling, AI technology can also enhance emergency response efforts during and after a disaster. For instance, drones equipped with AI algorithms can be deployed to assess the extent of damage in disaster-affected areas, identify survivors who may be trapped or injured, and provide real-time updates to rescue teams. This can help prioritize rescue operations, allocate resources efficiently, and save valuable time in the critical hours following a disaster. Moreover, AI can improve communication and coordination among different agencies involved in disaster response and recovery efforts. By creating a centralized platform to share information, allocate resources, and track progress, AI can streamline decision-making processes and ensure a more coordinated and efficient response. This is particularly crucial in large-scale disasters that require the involvement of multiple organizations, volunteers, and communities working together towards a common goal. In the aftermath of a disaster, AI can play a key role in optimizing the allocation of resources for recovery and reconstruction efforts. By analyzing data on infrastructure damage, population displacement, and economic losses, AI systems can recommend strategies for rebuilding communities, restoring critical services, and supporting vulnerable populations. AI can also facilitate access to financial assistance, insurance claims, and other support mechanisms for individuals and businesses affected by disasters. In general, the integration of AI technology into disaster management processes has the potential to revolutionize how we prepare for, respond to, and recover from natural disasters. By harnessing the power of AI to analyze data, generate insights, and facilitate decision-making, we can enhance the resilience of communities, reduce the impact of disasters, and ultimately save lives. Also, AI technology offers a powerful tool for improving disaster management efforts by providing predictive analytics, real-time insights, and efficient resource allocation. By leveraging AI capabilities, disaster management agencies can strengthen their capacity to anticipate, respond to, and recover from natural disasters, ultimately enhancing the safety and well-being of communities around the world.

7 | THE LIMITATIONS OF USE OF AI FOR SUSTAINABLE URBANIZATION

The growth of cities in the 21st century has been rapid, leading to increased pressure on resources, infrastructure, and the environment.

With urban populations projected to reach 68% by 2050, it has become crucial to find innovative solutions to ensure sustainable urbanization. AI has emerged as a potential tool to address these challenges, but there are limitations to its use in this context.

One major limitation of using AI for sustainable urbanization is the lack of inclusivity in AI technologies. AI systems are usually developed by a narrow group of individuals with limited perspectives and biases. This can result in AI systems that do not adequately address the needs and concerns of diverse populations in urban areas. For instance, AI algorithms used for traffic management may inadvertently favor certain neighborhoods over others, leading to increased congestion and pollution in marginalized communities. Moreover, the data used to train AI systems is often biased and incomplete. This can lead to AI systems that perpetuate existing inequalities and discrimination in urban areas. For example, AI systems used to predict crime rates may disproportionately target minority communities, leading to increased policing and surveillance in these areas. This can further deepen social tensions and marginalize already vulnerable populations.

Another limitation of using AI for sustainable urbanization is the lack of transparency and accountability in AI systems. Many AI algorithms operate as black-box systems, making it challenging to understand how they arrive at their decisions. This opacity can make it difficult to identify and rectify biases in AI systems, potentially causing harm to individuals and communities. For instance, AI systems used for predictive policing may unfairly target certain groups based on biased data, resulting in increased rates of incarceration and perpetuating cycles of poverty and crime. Furthermore, there are significant ethical concerns surrounding the use of AI for sustainable urbanization, particularly related to privacy and data security. AI systems collect vast amounts of data on individuals, including their movements, behaviors, and preferences. This data can be vulnerable to hacking and misuse, potentially leading to violations of privacy and civil liberties. For example, AI systems used for smart city initiatives may inadvertently expose sensitive information about individuals, such as their health status or financial situation, risking breaches of privacy and confidentiality. In addition, the use of AI for sustainable urbanization raises concerns about job displacement and economic inequality. AI technologies have the capacity to automate tasks traditionally performed by humans, leading to job loss in certain sectors. This can exacerbate existing inequalities in urban areas, as marginalized populations may lack access to retraining and reskilling opportunities. For example, the widespread adoption of AI in transportation and logistics may result in job loss for truck drivers and delivery workers, who may struggle to find alternative employment opportunities.

Moreover, the use of AI for sustainable urbanization poses regulatory challenges related to accountability and liability. In cases of malfunction or error in an AI system, it can be difficult to determine who is responsible for the resulting harm. This can lead to legal disputes and costly litigation, further impeding the implementation of AI technologies in urban areas. For example, in the event of a traffic accident involving an autonomous vehicle, it may be unclear whether the manufacturer, programmer, or operator is liable for the damages incurred.

Despite these limitations, there are opportunities to overcome the challenges of using AI for sustainable urbanization. One potential solution is to enhance transparency and accountability in AI systems through the development of ethical guidelines and regulatory frameworks. This can help ensure that AI technologies are developed and deployed in a responsible and equitable manner, benefiting all members of society. Another potential solution is to prioritize inclusivity and diversity in the development of AI technologies. By incorporating diverse perspectives and experiences into the design and implementation of AI systems, developers can create more inclusive and equitable solutions for sustainable urbanization. For example, AI algorithms used for affordable housing initiatives can be designed to prioritize the needs of low-income and marginalized communities, ensuring that housing resources are allocated fairly and equitably. In addition, stakeholders can collaborate to address the biases and limitations of AI technologies through ongoing monitoring and evaluation. By continuously assessing the impact of AI systems on individuals and communities, policymakers can identify and rectify potential harms before they escalate. For example, if an AI algorithm is found to disproportionately target certain groups, policymakers can adjust the algorithm to account for these biases and ensure fair and unbiased outcomes. In general, while AI holds great promise for revolutionizing sustainable urbanization, there are limitations to its use that must be addressed. By prioritizing inclusivity, transparency, and accountability in the development and deployment of AI technologies, stakeholders can overcome these challenges and create more equitable and sustainable cities for all residents. Through continued collaboration and dialog, we can harness the power of AI to build a more sustainable and resilient urban future.

8 | CONCLUSIONS

This article explores the different ways in which AI is being used to advance sustainable urban development. AI is notably impactful in urban planning by assisting in decision-making through data analysis on zoning, infrastructure development, and land use. This can result in better space utilization, decreased traffic congestion, and improved resource allocation. AI is also beneficial in resource and traffic management for sustainable urbanization. By monitoring traffic patterns and analyzing data, city officials can coordinate transportation systems more effectively, reduce emissions, and ease congestion. Predicting demand and adjusting supply helps optimize resource usage, leading to less waste, lower energy consumption, and more sustainable practices. Additionally, AI technology is crucial for monitoring air quality by analyzing sensor data to provide real-time insights and pinpoint sources of contamination. Targeted interventions based on this information can improve air quality and safeguard public health. Energy efficiency is another key focus where AI can make a difference in sustainable urbanization. By using data analytics and machine learning, AI systems can identify energy-saving opportunities, optimize usage, and incorporate renewable energy sources into the grid. This can lower carbon footprints, reduce energy costs, and encourage sustainable energy practices. Moreover, AI technology is instrumental in disaster management by predicting risks, identifying natural disasters, and aiding emergency responses. These capabilities help cities prepare for and respond to disasters effectively, minimizing damage and saving lives.

As a main result, AI offers innovative solutions to complex challenges, making cities more environmentally-friendly and efficient. From urban planning to disaster management, AI technology is shaping smarter, more sustainable cities for future generations.

AUTHOR CONTRIBUTIONS

All authors (MA) of the article are responsible to the design and implementation of the research to the analysis of the results and to the writing of the manuscript.

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All data generated or analyzed during this study are included in this published article.

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2

A city is not a computer

Shannon Mattern

Editor's introduction

"We want to build a city" is the strapline on a Y Combinator Research project web page titled "New Cities". Meanwhile Sidewalk Labs, an Alphabet company, sets out its challenge to respond to the following question – "What would a city look like if you started from scratch in the internet era—if you built a city 'from the internet up?'".[1] These tech visions set out the city as the ultimate Silicon Valley "start up". In this chapter, Mattern builds a case for challenging the rhetoric of this sort of computational thinking on cities.[2] Drawing on the marketing strategies of tech funders such as Y Combinator and Alphabet that position a city as the next challenge for the technological optimization model, she highlights the flaws in taking this approach to city-making. The broader argument draws on multiple readings which have revealed the problems with seeking to address complex socio-economic urban challenges with programmable solutions (Gabrys, 2014; Greenfield, 2013; Kitchin, 2014; Marvin et al., 2016; Vanolo, 2016). This ranges from the infrastructural, where a city embedded with networked sensors is reduced to a complex dataset down to the scale of the everyday where city inhabitants and their interactions are considered as users of technology with little or no agency. Mattern argues in this chapter that this focus on treating cities as complex problems to be solved by code means that we have lost a "critical perspective on how urban data become meaningful spatial information or translate into place-based knowledge".

Central to Mattern's discussion is the need to recognize the shortcomings in computational and data-driven models that presume an objectivity in urban data and fail to recognize or accommodate the range of critical and more importantly ethical decisions to the machine (Mattern, 2016, 2017). From IBM's original strategic move into the Smarter Cities arena to the more recent Sidewalk Labs' Toronto Waterfront project, technology companies have failed to adequately acknowledge and address the fact that technology is not neutral and is culturally, socially, economically and politically situated. By conveniently sidestepping the political implications of a city governed by data, schemes such as Sidewalk

Labs' Toronto Waterfront have failed to give adequate space to how the project might impact the relationship between local government and residents. Mattern makes a powerful case for demonstrating that treating the city as a problem to be solved through optimization is inherently a rewriting of any urban governance model, and therefore needs to be scrutinized on this level and not in terms of key performance indicators and productivity gains.

The dream of informatic urbanism is one underpinned by urban science approaches and propagated by tech companies, whilst the broader challenge is what role urban planners, designers and city inhabitants have in a city "designed from the internet up". Fundamentally, the city as a computer project is one where data analysists, marketeers and software engineers are the new planners and designers. As Mattern argues below, the processes of city-making are more complicated than simply rewriting code, and is challenged by "technologists (and political actors) who speak as if they could reduce urban planning to algorithms".

In the current urgent crisis of climate breakdown, it's widely acknowledged that we need to think radically about our cities. Treating the city as a computer may create a slick and palatable solution to city marketing programmes and looks compelling as a strapline on a website. Yet, as Mattern demonstrates, urban intelligence is more than data capture, feedback and processing, it is a much broader kind of knowledge that lives within bodies, minds and communities. This chapter thoughtfully and comprehensively captures the range of issues that combine to form a rejection of the technocractic, data driven vision of problem-solving in cities.

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Introduction

"What should a city optimize for?" Even in the age of peak Silicon Valley, that's a hard question to take seriously. (Hecklers on Twitter had a few ideas, like "fish tacos" and "pez dispensers.") Look past the sarcasm, though, and you'll find an ideology on the rise. The question was posed last summer[3] by Y Combinator—the formidable tech accelerator that has hatched a thousand startups, from AirBnB and Dropbox to robotic greenhouses and wine-by-the-glass delivery—as the entrepreneurs announced a new research agenda: building cities from scratch. *Wired's* verdict: "Not Actually Crazy" (Rhodes, 2016). Which is not to say wise. For every reasonable question Y Combinator asked—"How can cities help more of their residents be happy and reach their potential?"—there was a preposterous one: "How should we measure the effectiveness of a city (what are its KPIs)?" That's key performance indicators, for those not steeped in business intelligence jargon. There was hardly any mention of the urban designers, planners, and scholars who have been asking the big questions for centuries: "How do cities function, and how can they function better?"

Of course, it's possible that no city will be harmed in the making of this research. Half a year later, the public output of the New Cities project[4] consists of two blog posts, one announcing the program and the other reporting the first hire. Still, the rhetoric deserves close attention, because, frankly, in this new political age, all rhetoric demands scrutiny. At the highest levels of government, we see evidence and quantitative data manipulated or manufactured to justify reckless orders, disrupting not only "politics as usual," but also fundamental democratic principles. Much of the work in urban tech has the potential to play right into this new mode of governance.

Tech companies have come out forcefully against the Muslim travel ban, but where will they stand on subtler questions of social "optimization"? Autonomous vehicles and pervasive cameras and sensors are just the sort of disruptive technologies that an infrastructure-championing president might deem "tremendous." Donald Trump's chief strategist (who, years ago, ran the Biosphere 2 experiment into the ground) is also on the board of a data mining and analytics firm that seeks government contracts. Will the president start tweeting about how crime-ridden (and racialized) "inner cities" would be a whole lot better if they were run like computers?

It's a politically complicated environment, to say the least. Into the ring steps the first hire at New Cities: Ben Huh, founder of the meme-and-cat-pic empire Cheezburger. "There's no shortage of space to build new cities," he effervesced, in a post explaining his decision to join the Y Combinator project. "Technology can seed fertile starting conditions across nations and geographies." His goal for the six-month research position: to "create an open, repeatable system for rapid *cityforming* that maximize[s] human potential" (Huh, 2016). No pressure.

Meanwhile, Alphabet (née Google) is moving forward with plans to build its own optimized cities. Its urban-tech division, Sidewalk Labs, has already installed public WiFi kiosks on New York City streets: infrastructural nodes (known as "Links") that may someday exchange data with autonomous vehicles, public transit, and other urban systems.[5]

The company is also partnering with the U.S. Department of Transportation on efforts like the "Smart City Challenge," which awarded a US$50 million grant to Columbus, Ohio. Last June, on the same day Y Combinator announced its New Cities project, *The Guardian* published details of Alphabet's "Flow," the cloud software behind the mobility experiments in Columbus (Harris, 2016). Within months, partnerships were underway in 16 other cities (Davis, 2016).

Urban transportation is the first target for disruption, but it won't end there. Dan Doctoroff, the Michael Bloomberg associate who founded Sidewalk Labs, wonders, "What would a city look like if you started from scratch in the internet era—if you built a city 'from the internet up?'" In November, the company took another step in that direction, launching four new "labs" that will work on housing affordability, health care and social services, municipal processes, and community collaboration. The company plans to run pilot projects in select urban districts, then scale up. Announcing the expansion, Doctoroff recalled past "revolutions" in urban technologies:

> Looking at history, one can make the argument that the greatest periods of economic growth and productivity have occurred when we have integrated innovation into the physical environment, especially in cities. The steam engine, electricity grid, and automobile all fundamentally transformed urban life, but we haven't really seen much change in our cities since before World War II. If you compare pictures of cities from 1870 to 1940, it's like night and day. If you make the same comparison from 1940 to today, hardly anything has changed. Thus it's not surprising that, despite the rise of computers and the internet, growth has slowed and productivity increases are so low … So our mission is to accelerate the process of urban innovation.
>
> *(2016, n.p.)*

While Doctoroff has been telling some version of this story since Sidewalk Labs launched in 2015, the timing of the new expansion, three weeks after the U.S. presidential election, alters the context. As everyone was watching the drama at Trump Tower, the world's largest searching-mapping-driving-advertising-information-organizing company was throwing its resources behind a "fourth revolution" in urban infrastructure.

Dreams of an informatic urbanism

Of course, major companies like Alphabet have already dramatically reshaped the cities where they are headquartered,[6] but they have not yet had the luxury of building on a blank slate. The idea of the "new city" certainly isn't new, and the model now emerging in the United States has precedents in Asian and Middle Eastern countries, where Cisco, Siemens, and IBM have partnered with real-estate developers and governments to build "smart cities" *tabula rasa*.

We don't know how these urban experiments will fare. Since they are in a constant state of development, always "versioning" toward an optimized model ever on the horizon, they are not easily evaluated or critiqued (Halpern et al., 2017). If you believe the marketing hype, though, we're on the cusp of an urban future in which embedded sensors, ubiquitous cameras and beacons, networked smartphones, and the operating systems that link them all together, will produce unprecedented efficiency, connectivity, and social harmony. We're transforming the idealized topology of the open web and Internet of Things into urban form.

Programmer and tech writer Paul McFedries explains this thinking:

> The city is a computer, the streetscape is the interface, you are the cursor, and your smartphone is the input device. This is the user-based, bottom-up version of the city-as-computer idea, but there's also a top-down version, which is systems-based. It looks

at urban systems such as transit, garbage, and water and wonders whether the city could be more efficient and better organized if these systems were "smart."

(2014, p.36)

While projects like Sidewalk Labs and Y Combinator's New Cities were conceived in an age of big data and cloud computing, they are rooted in earlier reveries. Ever since the internet was little more than a few linked nodes, urbanists, technologists, and sci-fi writers have envisioned cybercities and e-topias built "from the 'net up'" (Boyer, 1995; Castells, 1989; Gibson, 1995; Mitchell, 2000, 1995). Modernist designers and futurists saw morphological parallels between urban forms and circuit boards. Just as new modes of telecommunication have always reshaped physical terrains and political economies, new computational methods have informed urban planning, modeling, and administration (Graham and Marvin, 1996; Light, 2004; Vallianatos, 2015).

Modernity is good at renewing metaphors, from the city as machine, to the city as organism or ecology, to the city as cyborgian merger of the technological and the organic.[7] Our current paradigm, the *city as computer*, appeals because it frames the messiness of urban life as programmable and subject to rational order. Anthropologist Hannah Knox explains, "As technical solutions to social problems, information and communications technologies encapsulate the promise of order over disarray … as a path to an emancipatory politics of modernity" (Knox, 2010, pp.187–188). And there are echoes of the pre-modern, too. The computational city draws power from an urban imaginary that goes back millennia, to the city as an apparatus for record-keeping and information management.

We've long conceived of our cities as knowledge repositories and data processors, and they've always functioned as such. Lewis Mumford observed that when the wandering rulers of the European Middle Ages settled in capital cities, they installed a "regiment of clerks and permanent officials" and established all manner of paperwork and policies (deeds, tax records, passports, fines, regulations), which necessitated a new urban apparatus, the office building, to house its bureaus and bureaucracy (Mumford, 1961, p.344). The classic example is the Uffizi (Offices) in Florence, designed by Giorgio Vasari in the mid-16th century, which provided an architectural template copied in cities around the world. "The repetitions and regimentations of the bureaucratic system"—the work of data processing, formatting, and storage—left a "deep mark," as Mumford put it, on the early modern city (Kittler, 1996, pp.721–722).

Yet the city's informational role began even earlier than that. Writing and urbanization developed concurrently in the ancient world, and those early scripts—on clay tablets, mudbrick walls, and landforms of various types—were used to record transactions, mark territory, celebrate ritual, and embed contextual information in landscape (Mattern, 2016b). Mumford described the city as a fundamentally communicative space, rich in information:

> Through its concentration of physical and cultural power, the city heightened the tempo of human intercourse and translated its products into forms that could be stored and reproduced. Through its monuments, written records, and orderly habits of association, the city enlarged the scope of all human activities, extending them backwards and forwards in time. By means of its storage facilities (buildings, vaults, archives, monuments, tablets, books), the city became capable of transmitting a complex culture from generation to generation, for it marshaled together not only the physical means but the human agents needed to pass on and enlarge this heritage. That remains the greatest of the city's gifts.

> As compared with the complex human order of the city, our present ingenious electronic mechanisms for storing and transmitting information are crude and limited.
>
> *(Mumford, 1961, p.569)*

Mumford's city is an assemblage of media forms (vaults, archives, monuments, physical and electronic records, oral histories, lived cultural heritage), agents (architectures, institutions, media technologies, people), and functions (storage, processing, transmission, reproduction, contextualization, operationalization).[8] It is a large, complex, and varied epistemological and bureaucratic apparatus. It is an information processor, to be sure, but it is also more than that.

Were he alive today, Mumford would reject the creeping notion that the city is simply the internet writ large. He would remind us that the processes of city-making are more complicated than writing parameters for rapid spatial optimization. He would inject history and happenstance. *The city is not a computer.* This seems an obvious truth, but it is being challenged now (again) by technologists (and political actors) who speak as if they could reduce urban planning to algorithms (for more on the algorithm as a timely conceptual model, see Mazzotti, 2017).

Why should we care about debunking obviously false metaphors? It matters because the metaphors give rise to technical models, which inform design processes, which in turn shape knowledges and politics, not to mention material cities. The sites and systems where we locate the city's informational functions—the places where we see information-processing, storage, and transmission "happening" in the urban landscape—shape larger understandings of urban intelligence.

Informational ecologies of the city

The idea of the city as an information-processing machine has in recent years manifested as a cultural obsession with urban sites of data storage and transmission. Scholars, artists, and designers write books, conduct walking tours, and make maps of internet infrastructures. We take pleasure in pointing at nondescript buildings that hold thousands of whirring servers, at surveillance cameras, camouflaged antennae, and hovering drones. We declare: "the city's computation happens here" (Mattern, 2013, 2016d).[9]

Yet such work runs the risk of reifying and essentializing information, even depoliticizing it. When we treat data as a "given" (which is, in fact, the etymology of the word), we see it in the abstract, as an urban fixture like traffic or crowds. We need to shift our gaze and look at data in context, at the lifecycle of urban information, distributed within a varied ecology of urban sites and subjects who interact with it in multiple ways. We need to see data's human, institutional, and technological creators, its curators, its preservers, its owners and brokers, its "users," its hackers and critics. As Mumford understood, there is more than information *processing* going on here. Urban information is *made*, commodified, accessed, secreted, politicized, and operationalized.

But where? Can we point to the chips and drives, cables and warehouses—the specific urban architectures and infrastructures—where this expanded ecology of information management resides and operates? I've written about the challenges of reducing complicated technical and intellectual structures to their material, geographic manifestations, i.e., mapping "where the data live" (Mattern, 2016d) (see also Amoore, 2018). Yet such exercises can be useful in identifying points of entry to the larger system. It's not only the infrastructural object that matters; it's also the personnel and paperwork and protocols, the machines and management practices, the conduits and cultural variables that shape terrain within the larger ecology of urban information.

So the next time you're staring up at a Domain Awareness camera, ask how it got there, how it generates data—not only how the equipment operates technically, but also what information it claims to be harvesting, and through what methodology—and whose interests it serves. And don't let the totalizing idea of the *city as computer* blind you to the countless other forms of data and sites of intelligence-generation in the city: municipal agencies and departments, universities, hospitals, laboratories, corporations. Each of these sites has a distinctive orientation toward urban intelligence. Let us consider a few of the more public ones.

First, the municipal archive. Most cities today have archives that contains records of administrative activity, finances, land ownership and taxes, legislation and labor. The archives of ancient Mesopotamian and Egyptian cities held similar material, although historians debate whether ancient record-keeping practices served similar documentary functions (O'Toole, 2004). Archives ensure financial accountability, symbolically legitimize governing bodies and colonial rulers, and erase the heritage of previous regimes and conquered populations. They monumentalize a culture's historical consciousness and intellectual riches. In the modern age, they also support scholarship (Walsham, 2016). Thus, the "information" inherent in the archive resides not solely in the content of its documents, but also in their very existence, their provenance and organization (there's much to be learned about the ideals of a culture by examining its archival forms), and even in the archive's omissions and erasures (Stoler, 2010).

Of course, not all archives are ideologically equal. Community archives validate the personal histories and intellectual contributions of diverse publics. Meanwhile, law enforcement agencies and customs and immigration offices are networked with geographically distributed National Security Agency repositories and other federal black boxes. These archives are not of the same species, nor do they "process" "data" in the same fashion.

Practices and politics of curation and access have historically distinguished archives from another key site of urban information: libraries. Whereas archives collect unpublished materials and attend primarily to their preservation and security, libraries collect published materials and aim to make them intelligible and accessible to patrons. In practice, such distinctions are fuzzy and contested, especially today, as many archives seek to be more public-facing. Nevertheless, these two institutions embody different knowledge regimes and ideologies.

Modern libraries and librarians have sought to empower patrons to access information across platforms and formats, and to critically assess bias, privacy, and other issues under the rubric of "information literacy" (Mattern, 2016a). They build a critical framework around their resources, often in partnership with schools and universities. Further, libraries perform vital symbolic functions, embodying the city's commitment to its intellectual heritage (which may include heritage commandeered through imperial activities).

Similarly, the city's museums reflect its commitment to knowledge in embodied form, to its artifacts and material culture. Again, such institutions are open to ideological critique. Acquisition policies, display practices, and access protocols are immediate and tangible, and they reflect particular cultural and intellectual politics.

Just as important as the data stored and accessed on city servers, in archival boxes, on library shelves and museum walls are the forms of urban intelligence that cannot be easily contained, framed, and catalogued. We need to ask: What place-based "information" doesn't fit on a shelf or in a database? What are the non-textual, un-recordable forms of cultural memory? These questions are especially relevant for marginalized populations, indigenous cultures, and developing nations. Performance studies scholar Diana Taylor urges us to acknowledge ephemeral, performative forms of knowledge, such as dance, ritual, cooking, sports, and speech (Taylor, 2003). These forms cannot be reduced to "information," nor can they be "processed," stored, or transmitted via fiber-optic cable. Yet they are vital urban intelligences that live within bodies, minds, and communities.

Finally, consider data of the environmental, ambient, "immanent" kind. Malcolm McCullough has shown that our cities are full of fixed architectures, persistent terrains, and reliable environmental patterns that anchor all the unstructured data and image streams that float on top (McCullough, 2014, p.36, 42). What can we learn from the "nonsemantic information" inherent in shadows, wind, rust, in the signs of wear on a well-trodden staircase, the creaks of a battered bridge—all the indexical messages of our material environments? I'd argue that the intellectual value of this ambient, immanent information exceeds its function as stable ground for the city's digital flux. Environmental data are just as much figure as they are ground. They remind us of necessary truths: that urban intelligence comes in multiple forms, that it is produced within environmental as well as cultural contexts, that it is reshaped over the *longue durée* by elemental exposure and urban development, that it can be lost or forgotten. These data remind us to think on a climatic scale, a geologic scale, as opposed to the scale of financial markets, transit patterns, and news cycles.

Here's some geologic insight from T. S. Eliot's 1934 poem "The Rock":

> Where is the Life we have lost in living?
> Where is the wisdom we have lost in knowledge?
> Where is the knowledge we have lost in the information?
>
> *Eliot, T. S. (1934)*

Management theorist Russell Ackoff took Eliot's idea one step further, proposing the now famous (and widely debated) hierarchy: Data < Information < Knowledge < Wisdom (Sharma, 2008; Weinberger, 2010). Each level of processing implies an extraction of utility from the level before. Thus, contextualized or patterned data can be called information. Or, to quote philosopher and computer scientist Frederick Thompson, information is "a product that results from applying the processes of organization to the raw material of experience, much like steel is obtained from iron ore." Swapping the industrial metaphor for an artistic one, he writes, "data are to the scientist like the colors on the palette of the painter. It is by the artistry of his theories that we are informed. It is the organization that is the information".[10] Thompson's mixed metaphors suggest that there are multiple ways of turning data into information and knowledge into wisdom.

Yet the term "information processing," whether employed within computer science, cognitive psychology, or urban design, typically refers to *computational* methods. As Riccardo Manzotti explains, when neuroscientists adopt the metaphor of the *brain as computer*, they imply that information is "stuff" that's mentally "processed," which they know is not true in any real sense. The metaphor survives because it makes an irresistible claim about "how marvelously complex we are and how clever scientists have become" (Manzotti and Parks, 2016). Psychologist Robert Epstein laments that "some of the world's most influential thinkers have made grand predictions about humanity's future that depend on the validity of the metaphor" (Epstein, 2016). But the appeal of analogy is nothing new. Throughout history, the brain (like the city) has been subjected to bad metaphors derived from the technologies of the time. According to Epstein, we've imagined ourselves as lumps of clay infused with spirits, as hydraulic or electro-chemical systems, as automata. The *brain as* *computer* is just the latest link in a long chain of metaphors that powerfully shape scientific endeavor in their own images.

The *city as computer* model likewise conditions urban design, planning, policy, and administration—even residents' everyday experience—in ways that hinder the development of healthy, just, and resilient cities. Let's apply Manzotti's and Epstein's critiques at the city scale. We have seen that urban ecologies "process" data by means that are not strictly algorithmic, and that not all urban intelligences can be called "information." One can't "process" the local cultural effects of long-term weather patterns or derive insights from the generational evolution of a neighborhood without a degree of sensitivity that exceeds mere computation. Urban intelligence of this kind involves site-based experience, participant observation, sensory engagement. We need new models for thinking about cities that *do not compute*, and we need new terminology. In contemporary urban discourses, where "data" rhetoric is often frothy and fetishistic, we seem to have lost critical perspective on how urban data become meaningful spatial information or translate into place-based knowledge.

We need to expand our *repertoire* (to borrow a term from Diana Taylor) of urban intelligences, to draw upon the wisdom of information scientists and theorists, archivists, librarians, intellectual historians, cognitive scientists, philosophers, and others who think about the management of information and the production of knowledge (Foth et al., 2007). They can help us better understand the breadth of intelligences that are integrated within our cities, which would be greatly impoverished if they were to be rebuilt, or built anew, with computational logic as their prevailing epistemology.

We could also be better attuned to the lifecycles of urban information resources—to their creation, curation, provision, preservation, and destruction—and to the assemblages of urban sites and subjects that make up our cities' intellectual ecologies. "If we think of the city as a long-term construct, with more complex behaviors and processes of formation, feedback, and processing," architect Tom Verebes proposes, then we can imagine it as an organization, or even an organism, that can learn (Verebes, 2016). Urbanists and designers are already drawing on concepts and methods from artificial intelligence research: neural nets, cellular processes, evolutionary algorithms, mutation and evolution.[11] Perhaps quantum entanglement and other computer science breakthroughs could reshape the way we think about urban information, too. Yet we must be cautious to avoid translating this interdisciplinary intelligence into a new urban formalism.

Instead of more gratuitous parametric modeling, we need to think about urban epistemologies that embrace memory and history; that recognize spatial intelligence as sensory and experiential; that consider other species' ways of knowing; that appreciate the wisdom of local crowds and communities; that acknowledge the information embedded in the city's facades, flora, statuary, and stairways; that aim to integrate forms of distributed cognition paralleling our brains' own distributed cognitive processes.

We must also recognize the shortcomings in models that presume the objectivity of urban data and conveniently delegate critical, often ethical decisions to the machine. We, humans, *make* urban information by various means: through sensory experience, through long-term exposure to a place, and, yes, by systematically filtering data. It's essential to make space in our cities for those diverse methods of knowledge production. And we have to grapple with the political and ethical implications of our methods and models, embedded in all acts of planning and design. *City-making* is always, simultaneously, an enactment of *city-knowing*—which cannot be reduced to computation.

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Notes

1 Doctoroff (2016) cited in https://medium.com/sidewalk-talk/reimagining-cities-from-the-internet-up-5923d6be63ba
2 Originally published in *Places Journal* in 2017.
3 See Cheung andAltman (2016) The post drew responses on Twitter from designer and urbanist Fred Scharmen ("fish tacos") and visual journalist Erik Reyna ("pez dispensers"), among others.
4 https://cities.ycr.org/
5 Sidewalk Labs is a key investor in Intersection, the "municipal media company" that is a partner in LinkNYC. See Brown, E. (2016), Mattern, S. (2016c), Lessin, (2016), and Weinberg (2016).
6 See Susie Cagle, "Why One Silicon Valley City Said 'No' to Google," *Next City*, May 11, 2015; Sean Hollister, "Welcome to Googletown," *The Verge*, February 26, 2014; Chris Morris-Lent,"How Amazon Swallowed Seattle," *Gawker*, August 18, 2015.
7 Some argue that the city-as-machine has a much deeper history, as evidenced by use of grid layouts, linear patterns, and regular geometric forms since ancient times, and by the use of standardized patterns for colonial urban development. See, for instance, Kevin Lynch, *Good City Form* (Cambridge, MA: MIT Press, 1981): 81–88. See also Matthew Gandy, "Cyborg Urbanization: Complexity and Monstrosity in the Contemporary City," *International Journal of Urban and Regional Research* 29: (March 2005): 26–49, https://doi.org/10.1111/j.1468–2427.2005.00568.x; Peter Nientied, "Metaphor and Urban Studies: A Crossover, Theory and a Case Study of SS Rotterdam," *City, Territory and Architecture* 3:21 (2016), https://doi.org/10.1186/s40410-016-0051-z; William Solesbury, "How Metaphors Help Us Understand Cities," *Geography* 99:3 (Autumn 2014): 139–42; Tom Verebes (2016).
8 Marcus Foth's conception of "urban informatics" is similarly capacious: it encompasses "the collection, classification, storage, retrieval, and dissemination of recorded knowledge," either (1) in a city or (2) "of, relating to, characteristic of, or constituting a city." See Foth, M. (ed.) *Handbook of Research on Urban Informatics: The Practice and Promise of the Real-Time City* (Hershey, PA: Information Science Reference, 2009): xxiii. Such a definition acknowledges a wide variety of informational functions, contents, and contexts. Yet his focus on recorded knowledge, and on informatics' reputation as a "science" of data processing, still limits our understanding of the city's epistemological functions.
9 For prominent examples, see Andrew Blum, *Tubes: A Journey to the Center of the Internet* (New York: HarperCollins, 2012), and the work of Ingrid Burrington and Mél Hogan.
10 Quoted in Marcia J. Bates, "Information," in Marcia J. Bates, Mary Niles Maac, eds., Encyclopedia of Library and Information Sciences, 3rd ed. (New York: CRC Press, 2010): 2347–2360, available online https://pages.gseis.ucla.edu/faculty/bates/articles/information.html". See also Rafael Capurro and Birger Hjørland, "The Concept of Information," in Blaise Cronin (ed.), *The Annual Review of Information Science and Technology*, Vol 37 (2003): 343–411.
11 See, for instance, the work of Michael Batty.

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